

CITY of MALABON UNIVERSITY

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SIA101 - System Integration and Architecture 1	
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Activity 1: Service Oriented Architecture (SOA)	

As an IT consultant for City of Malabon University (CMU), I see one big problem: CMU has separate systems for enrollment, library, accounting, and student records that do not communicate. What is Service-Oriented Architecture (SOA)? SOA is a way of building systems as small, reusable services that share data through standard interfaces, so different applications can work together without being rebuilt.

Who are the key roles in SOA? The Service Provider offers a function, such as “check student balance.” The Service Consumer uses that function, like the enrollment system checking if a student can enroll. The Service Registry is a directory where providers list their services and consumers find them, like a phone book for services.

Which SOA principles would improve CMU’s system? First, loose coupling: services can change without breaking others, so updating the library will not stop enrollment. Second, reusability: the same “verify student ID” service can be used by the library, accounting, and records offices. Third, standardization: all services use common formats and rules, which reduces errors and training time. Fourth, discoverability: services are listed in the registry, so new apps can find and use them quickly.

How would these principles fix CMU’s current issues? Today, staff re-enter data and students wait while offices call each other. With SOA, when a student pays fees, the accounting service updates the balance, and the enrollment service reads it at once. The library can check the same record to allow or block borrowing. The student records office can pull grades and units without manual exports.

What benefits will CMU gain after implementing SOA? Students will enroll faster and with fewer errors. Staff will spend less time on manual tasks and more time on support. IT will deploy updates safely, because services are isolated. The university will save money by reusing services instead of building new integrations each time. In short, SOA turns CMU’s siloed systems into a connected campus platform that is easier to run, cheaper to maintain, and simpler to improve.